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Studierichting: Informatica

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$$\begin{aligned} \int_{-5}^{-1} x\sqrt{x+5}dx &\stackrel{t=\sqrt{x+5}}{=} \int_0^2 (t^2-5) t \cdot 2tdt = 2 \int_0^2 t^4 dt - 10 \int_0^2 t^2 dt \\ &= 2 \left[\frac{t^5}{5} \right]_0^2 - 10 \left[\frac{t^3}{3} \right]_0^2 = \frac{2}{5} (32 - 0) - \frac{10}{3} (8 - 0) \\ &= \frac{64}{5} - \frac{80}{3} = -\frac{208}{15} \end{aligned}$$

References